



K.R. MANGALAM UNIVERSITY
THE COMPLETE WORLD OF EDUCATION

DBMS LAB FILE

DBMS CAPSTONE PROJECTS



STUDENT NAME : BADAL PRASAD



ROLL NUMBER : 2501351020



SUBJECT NAME : DATABASE MANAGEMENT
SYSTEM



FACULTY NAME : Dr. Preeti Rathi Mam



SUBMISSION DATE : 01-MAY-2026

GIT HUB LINK :





2. Hospital Management System	- Design a schema to manage patient records including medical history, appointments, and treatments.
	- Implement queries to track patient admissions, discharges, and transfers between departments.
	- Handle complex queries for scheduling surgeries, managing hospital resources, and ensuring patient safety.

1. Introduction

A **Hospital Management System (HMS)** is a database-based application designed to manage hospital operations efficiently. It helps in storing and organizing patient records, doctor details, appointments, medical history, and department data.

This system ensures:

- Efficient data storage
- Easy retrieval of patient records
- Better hospital resource management
- Improved patient care and safety

In this project, we use **SQL (Structured Query Language)** to design and manage a hospital database

2. Objective

The main objectives of this assignment are:

- To design a **relational database** for a hospital system
- To implement **tables with relationships (Primary Key & Foreign Key)**
- To perform **CRUD operations (Create, Insert, Retrieve)**
- To execute **complex SQL queries** such as:
 - Joins
 - Group By
 - Filtering (WHERE clause)
- To track:
 - Patient admissions & discharges
 - Doctor appointments
 - Department-wise patient records
- To ensure **data consistency and integrity**



3. Database Creation

|| SQL Query ||

-- || NAME : BADAL PRASAD ||
-- || ROLL NO. : 2501351020 ||
-- || COURSE : DBMS ||

-- || CREATE DATABASE ||

CREATE DATABASE IF NOT EXISTS HospitalDB;
USE HospitalDB;

-- || OUTPUT ||

```
mysql> show databases;
+-----+
| Database |
+-----+
| classtest |
| college_database |
| company |
| company_database |
| experiment_1 |
| hospitaldb |
| information_schema |
| mysql |
| performance_schema |
| sakila |
| sys |
| world |
+-----+
12 rows in set (0.03 sec)
```

|| OUTPUT TABLES ||

```
mysql> USE hospitaldb;
Database changed
mysql> SHOW TABLES;
+-----+
| Tables_in_hospitaldb |
+-----+
| admissions |
| appointments |
| departments |
| doctors |
| medical_history |
| patients |
+-----+
6 rows in set (0.02 sec)
```

-- || CREATE TABLE OF DOCTORS ||

CREATE TABLE Doctors (
 doctor_id INT PRIMARY KEY AUTO_INCREMENT,
 name VARCHAR(50),
 specialization VARCHAR(50)
);

-- || INSERT VALUE Doctors ||

```
mysql> USE hospitaldb;
Database changed
mysql> SHOW TABLES;
+-----+
| Tables_in_hospitaldb |
+-----+
| admissions |
| appointments |
| departments |
| doctors |
| medical_history |
| patients |
+-----+
```

|| OUTPUT TABLES ||

INSERT INTO Doctors (name, specialization) VALUES
('rajveer singh', 'Cardiologist'),
('BADAL PRASAD', 'Neurologist'),
('SHRUTI SINGH', 'Pediatrician'),
('SHIV', 'Orthopedic'),
('ADITYA', 'General Physician');

```
mysql> SELECT * FROM doctors;
+-----+-----+-----+
| doctor_id | name | specialization |
+-----+-----+-----+
| 1 | Rajveer Singh | Cardiologist |
| 2 | Badal Prasad | Neurologist |
| 3 | Shruti Singh | Pediatrician |
| 4 | Shiv | Orthopedic |
| 5 | Aditya | General Physician |
+-----+-----+-----+
5 rows in set (0.00 sec)
```




-- || CREATE TABLE OF Patients ||

```
CREATE TABLE Patients (  
  patient_id INT PRIMARY KEY AUTO_INCREMENT,  
  name VARCHAR(50),  
  age INT,  
  gender VARCHAR(10),  
  contact VARCHAR(15)  
);
```

```
mysql> USE hospitaldb;  
Database changed  
mysql> SHOW TABLES;  
+-----+  
| Tables_in_hospitaldb |  
+-----+  
| admissions            |  
| appointments          |  
| departments           |  
| doctors               |  
| medical_history        |  
| patients              |  
+-----+
```

-- || INSERT VALUE Patients ||

```
INSERT INTO Patients (name, age, gender, contact) VALUES  
( 'Aryan Goyal', 22, 'Male', '9000000001'),  
( 'Vansh Goel', 23, 'Male', '9000000002'),  
( 'Rashi Yadav', 21, 'Female', '9000000003'),  
( 'Saumya Barthwal', 24, 'Female', '9000000004'),  
( 'Abhay Mundepe', 25, 'Male', '9000000005'),  
( 'Sanjeev Sangwan', 30, 'Male', '9000000006'),  
( 'Harsh Panchal', 26, 'Male', '9000000007'),  
( 'Yuvraj', 20, 'Male', '9000000008'),  
( 'Krish', 19, 'Male', '9000000009'),  
( 'Devansh Thakur', 27, 'Male', '9000000010'),  
( 'Shailjanand', 28, 'Male', '9000000011'),  
( 'Manju', 35, 'Female', '9000000012'),  
( 'Tarun Yadav', 29, 'Male', '9000000013'),  
( 'Adit Ghosh', 23, 'Male', '9000000014'),  
( 'Vipul', 31, 'Male', '9000000015'),  
( 'Subhrajeet Dash', 32, 'Male', '9000000016');
```

|| OUTPUT TABLES ||

```
mysql> SELECT * FROM patients;  
+-----+-----+-----+-----+-----+  
| patient_id | name          | age | gender | contact |  
+-----+-----+-----+-----+-----+  
| 1          | Aryan Goyal  | 22  | Male   | 9000000001 |  
| 2          | Vansh Goel   | 23  | Male   | 9000000002 |  
| 3          | Rashi Yadav  | 21  | Female | 9000000003 |  
| 4          | Saumya Barthwal | 24  | Female | 9000000004 |  
| 5          | Abhay Mundepe | 25  | Male   | 9000000005 |  
| 6          | Sanjeev Sangwan | 30  | Male   | 9000000006 |  
| 7          | Harsh Panchal | 26  | Male   | 9000000007 |  
| 8          | Yuvraj       | 20  | Male   | 9000000008 |  
| 9          | Krish        | 19  | Male   | 9000000009 |  
| 10         | Devansh Thakur | 27  | Male   | 9000000010 |  
| 11         | Shailjanand  | 28  | Male   | 9000000011 |  
| 12         | Manju        | 35  | Female | 9000000012 |  
| 13         | Tarun Yadav  | 29  | Male   | 9000000013 |  
| 14         | Adit Ghosh   | 23  | Male   | 9000000014 |  
| 15         | Vipul        | 31  | Male   | 9000000015 |  
| 16         | Subhrajeet Dash | 32  | Male   | 9000000016 |  
+-----+-----+-----+-----+-----+  
16 rows in set (0.00 sec)
```

-- || CREATE TABLE OF Medical_History ||

```
CREATE TABLE Medical_History (  
  history_id INT PRIMARY KEY AUTO_INCREMENT,  
  patient_id INT,  
  disease VARCHAR(100),  
  treatment VARCHAR(100),  
  FOREIGN KEY (patient_id) REFERENCES Patients(patient_id)  
);
```

```
mysql> USE hospitaldb;  
Database changed  
mysql> SHOW TABLES;  
+-----+  
| Tables_in_hospitaldb |  
+-----+  
| admissions            |  
| appointments          |  
| departments           |  
| doctors               |  
| medical_history        |  
| patients              |  
+-----+
```

-- || INSERT VALUE Medical_History ||

```
INSERT INTO Medical_History (patient_id, disease, treatment) VALUES  
(1, 'Fever', 'Paracetamol'),  
(2, 'Migraine', 'Painkillers'),  
(3, 'Diabetes', 'Insulin'),  
(4, 'Asthma', 'Inhaler'),  
(5, 'Fracture', 'Surgery'),  
(6, 'Hypertension', 'Medication'),  
(7, 'Cold', 'Antibiotics'),  
(8, 'Allergy', 'Antihistamine'),  
(9, 'Flu', 'Rest and Fluids'),  
(10, 'Infection', 'Antibiotics');
```

|| OUTPUT TABLES ||

```
mysql> SELECT * FROM medical_history;  
+-----+-----+-----+-----+  
| history_id | patient_id | disease      | treatment |  
+-----+-----+-----+-----+  
| 1          | 1          | Fever       | Paracetamol |  
| 2          | 2          | Migraine    | Painkillers |  
| 3          | 3          | Diabetes    | Insulin     |  
| 4          | 4          | Asthma      | Inhaler     |  
| 5          | 5          | Fracture    | Surgery     |  
| 6          | 6          | Hypertension | Medication  |  
| 7          | 7          | Cold        | Antibiotics |  
| 8          | 8          | Allergy     | Antihistamine |  
| 9          | 9          | Flu         | Rest and Fluids |  
| 10         | 10         | Infection   | Antibiotics |  
+-----+-----+-----+-----+  
10 rows in set (0.01 sec)
```




-- || CREATE TABLE OF Appointments ||

```
CREATE TABLE Appointments (  
    appointment_id INT PRIMARY KEY AUTO_INCREMENT,  
    patient_id INT,  
    doctor_id INT,  
    appointment_date DATE,  
    status VARCHAR(20),  
    FOREIGN KEY (patient_id) REFERENCES Patients(patient_id),  
    FOREIGN KEY (doctor_id) REFERENCES Doctors(doctor_id)  
);
```

```
mysql> USE hospitaldb;  
Database changed  
mysql> SHOW TABLES;  
+-----+  
| Tables_in_hospitaldb |  
+-----+  
| admissions            |  
| appointments          |  
| departments           |  
| doctors               |  
| medical_history       |  
| patients              |  
+-----+
```

-- || INSERT VALUE Appointment ||

```
INSERT INTO Appointments (patient_id, doctor_id, appointment_date, status) VALUES  
(1, 1, '2026-05-03', 'Scheduled'),  
(2, 2, '2026-05-04', 'Completed'),  
(3, 3, '2026-05-05', 'Scheduled'),  
(4, 4, '2026-05-06', 'Completed'),  
(5, 5, '2026-05-07', 'Scheduled'),  
(6, 1, '2026-05-08', 'Completed'),  
(7, 2, '2026-05-09', 'Scheduled'),  
(8, 3, '2026-05-10', 'Completed'),  
(9, 4, '2026-05-11', 'Scheduled'),  
(10, 5, '2026-05-12', 'Completed');
```

|| OUTPUT TABLES ||

```
mysql> SELECT * FROM appointments;  
+-----+  
| appointment_id | patient_id | doctor_id | appointment_date | status |  
+-----+  
| 1              | 1          | 1         | 2026-05-03      | Scheduled |  
| 2              | 2          | 2         | 2026-05-04      | Completed |  
| 3              | 3          | 3         | 2026-05-05      | Scheduled |  
| 4              | 4          | 4         | 2026-05-06      | Completed |  
| 5              | 5          | 5         | 2026-05-07      | Scheduled |  
| 6              | 6          | 1         | 2026-05-08      | Completed |  
| 7              | 7          | 2         | 2026-05-09      | Scheduled |  
| 8              | 8          | 3         | 2026-05-10      | Completed |  
| 9              | 9          | 4         | 2026-05-11      | Scheduled |  
| 10             | 10         | 5         | 2026-05-12      | Completed |  
+-----+  
10 rows in set (0.00 sec)
```

-- || CREATE TABLE Departments ||

```
CREATE TABLE Departments (  
    dept_id INT PRIMARY KEY AUTO_INCREMENT,  
    dept_name VARCHAR(50)  
);
```

```
mysql> USE hospitaldb;  
Database changed  
mysql> SHOW TABLES;  
+-----+  
| Tables_in_hospitaldb |  
+-----+  
| admissions            |  
| appointments          |  
| departments           |  
| doctors               |  
| medical_history       |  
| patients              |  
+-----+
```

-- || INSTER VALUE DEPARTMENT ||

```
INSERT INTO Departments (dept_name) VALUES  
( 'Cardiology'),  
( 'Neurology'),  
( 'Pediatrics'),  
( 'Orthopedics'),  
( 'General Medicine'),  
( 'Dermatology'),  
( 'Oncology'),  
( 'Radiology'),  
( 'ENT'),  
( 'Gastroenterology');
```

|| OUTPUT TABLES ||

```
mysql> SELECT * FROM departments;  
+-----+  
| dept_id | dept_name |  
+-----+  
| 1       | Cardiology |  
| 2       | Neurology  |  
| 3       | Pediatrics |  
| 4       | Orthopedics |  
| 5       | General Medicine |  
| 6       | Dermatology |  
| 7       | Oncology   |  
| 8       | Radiology  |  
| 9       | ENT        |  
| 10      | Gastroenterology |  
+-----+  
10 rows in set (0.00 sec)
```




-- || CREATE TABLE Admissions ||

```
CREATE TABLE Admissions (  
    admission_id INT PRIMARY KEY AUTO_INCREMENT,  
    patient_id INT,  
    dept_id INT,  
    admission_date DATE,  
    discharge_date DATE,  
    FOREIGN KEY (patient_id) REFERENCES Patients(patient_id),  
    FOREIGN KEY (dept_id) REFERENCES Departments(dept_id)  
);
```

```
mysql> USE hospitaldb;  
Database changed  
mysql> SHOW TABLES;  
+-----+  
| Tables_in_hospitaldb |  
+-----+  
| admissions            |  
| appointments          |  
| departments           |  
| doctors               |  
| medical_history       |  
| patients              |  
+-----+
```

-- || INSERT VALUES ADMISSES ||

```
INSERT INTO Admissions (patient_id, dept_id, admission_date, discharge_date) VALUES  
(1, 1, '2026-05-01', NULL),  
(2, 2, '2026-05-02', '2026-05-05'),  
(3, 3, '2026-05-03', NULL),  
(4, 4, '2026-05-04', '2026-05-06'),  
(5, 5, '2026-05-05', NULL),  
(6, 6, '2026-05-06', NULL),  
(7, 7, '2026-05-07', '2026-05-10'),  
(8, 8, '2026-05-08', NULL),  
(9, 9, '2026-05-09', '2026-05-12'),  
(10, 10, '2026-05-10', NULL);
```

|| OUTPUT TABLES ||

```
mysql> SELECT * FROM Admissions;  
+-----+-----+-----+-----+-----+  
| admission_id | patient_id | dept_id | admission_date | discharge_date |  
+-----+-----+-----+-----+-----+  
| 1            | 1          | 1       | 2026-05-01     | NULL           |  
| 2            | 2          | 2       | 2026-05-02     | 2026-05-05    |  
| 3            | 3          | 3       | 2026-05-03     | NULL           |  
| 4            | 4          | 4       | 2026-05-04     | 2026-05-06    |  
| 5            | 5          | 5       | 2026-05-05     | NULL           |  
| 6            | 6          | 6       | 2026-05-06     | NULL           |  
| 7            | 7          | 7       | 2026-05-07     | 2026-05-10    |  
| 8            | 8          | 8       | 2026-05-08     | NULL           |  
| 9            | 9          | 9       | 2026-05-09     | 2026-05-12    |  
| 10           | 10         | 10      | 2026-05-10     | NULL           |  
+-----+-----+-----+-----+-----+  
10 rows in set (0.00 sec)
```

-- || Track Patient Admissions ||

```
SELECT p.name, d.dept_name, a.admission_date  
FROM Admissions a  
JOIN Patients p ON a.patient_id = p.patient_id  
JOIN Departments d ON a.dept_id = d.dept_id;
```

|| OUTPUT TABLES ||

```
mysql> SELECT p.name, d.dept_name, a.admission_date  
-> FROM Admissions a  
-> JOIN Patients p ON a.patient_id = p.patient_id  
-> JOIN Departments d ON a.dept_id = d.dept_id;  
+-----+-----+-----+  
| name      | dept_name | admission_date |  
+-----+-----+-----+  
| Aryan Goyal | Cardiology | 2026-05-01     |  
| Vansh Goel  | Neurology | 2026-05-02     |  
| Rashmi Yadav | Pediatrics | 2026-05-03     |  
| Saumya Barthwal | Orthopedics | 2026-05-04     |  
| Abhay Munde | General Medicine | 2026-05-05     |  
| Sanjeev Sangwan | Dermatology | 2026-05-06     |  
| Harsh Panchal | Oncology   | 2026-05-07     |  
| Yuvraj      | Radiology  | 2026-05-08     |  
| Krish       | ENT        | 2026-05-09     |  
| Devansh Thakur | Gastroenterology | 2026-05-10     |  
+-----+-----+-----+  
10 rows in set (0.02 sec)
```

-- || Track Discharges ||

```
SELECT p.name, a.discharge_date  
FROM Admissions a  
JOIN Patients p ON a.patient_id = p.patient_id  
WHERE a.discharge_date IS NOT NULL;
```

```
mysql> SELECT p.name, a.discharge_date  
-> FROM Admissions a  
-> JOIN Patients p ON a.patient_id = p.patient_id  
-> WHERE a.discharge_date IS NOT NULL;  
+-----+-----+  
| name      | discharge_date |  
+-----+-----+  
| Vansh Goel | 2026-05-05     |  
| Saumya Barthwal | 2026-05-06     |  
| Harsh Panchal | 2026-05-10     |  
| Krish      | 2026-05-12     |  
+-----+-----+  
4 rows in set (0.01 sec)
```

|| OUTPUT TABLES ||

-- || Track Transfers Between Departments ||

|| OUTPUT TABLES ||

```
SELECT p.name, d.dept_name  
FROM Admissions a  
JOIN Patients p ON a.patient_id = p.patient_id  
JOIN Departments d ON a.dept_id = d.dept_id  
ORDER BY p.name;
```

```
mysql> SELECT p.name, d.dept_name  
-> FROM Admissions a  
-> JOIN Patients p ON a.patient_id = p.patient_id  
-> JOIN Departments d ON a.dept_id = d.dept_id  
-> ORDER BY p.name;  
+-----+-----+  
| name      | dept_name |  
+-----+-----+  
| Abhay Munde | General Medicine |  
| Aryan Goyal | Cardiology      |  
| Devansh Thakur | Gastroenterology |  
| Harsh Panchal | Oncology        |  
| Krish       | ENT             |  
| Rashmi Yadav | Pediatrics      |  
| Sanjeev Sangwan | Dermatology     |  
| Saumya Barthwal | Orthopedics     |  
| Vansh Goel  | Neurology       |  
| Yuvraj      | Radiology       |  
+-----+-----+  
10 rows in set (0.02 sec)
```



-- || Surgery Scheduling ||

```
SELECT p.name, d.name AS doctor, a.appointment_date
FROM Appointments a
JOIN Patients p ON a.patient_id = p.patient_id
JOIN Doctors d ON a.doctor_id = d.doctor_id
WHERE a.status = 'Scheduled';
```

```
mysql> SELECT p.name, d.name AS doctor, a.appointment_date
-> FROM Appointments a
-> JOIN Patients p ON a.patient_id = p.patient_id
-> JOIN Doctors d ON a.doctor_id = d.doctor_id
-> WHERE a.status = 'Scheduled';
```

name	doctor	appointment_date
Aryan Goyal	Rajveer Singh	2026-05-03
Rashi Yadav	Shruti Singh	2026-05-05
Abhay Mundepi	Aditya	2026-05-07
Harsh Panchal	Badal Prasad	2026-05-09
Krish	Shiv	2026-05-11

5 rows in set (0.01 sec)

← || OUTPUT TABLES ||

-- || Hospital Resource Management (Department-wise Patients Count) ||

```
SELECT d.dept_name, COUNT(a.patient_id) AS total_patients
FROM Admissions a
JOIN Departments d ON a.dept_id = d.dept_id
GROUP BY d.dept_name;
```

```
mysql> SELECT d.dept_name, COUNT(a.patient_id) AS total_patients
-> FROM Admissions a
-> JOIN Departments d ON a.dept_id = d.dept_id
-> GROUP BY d.dept_name;
```

dept_name	total_patients
Cardiology	1
Neurology	1
Pediatrics	1
Orthopedics	1
General Medicine	1
Dermatology	1
Oncology	1
Radiology	1
ENT	1
Gastroenterology	1

10 rows in set (0.01 sec)

← || OUTPUT TABLES ||

-- || Patient Safety (Patients without Discharge) ||

```
SELECT p.name, a.admission_date
FROM Admissions a
JOIN Patients p ON a.patient_id = p.patient_id
WHERE a.discharge_date IS NULL;
```

```
mysql> SELECT p.name, a.admission_date
-> FROM Admissions a
-> JOIN Patients p ON a.patient_id = p.patient_id
-> WHERE a.discharge_date IS NULL;
```

name	admission_date
Aryan Goyal	2026-05-01
Rashi Yadav	2026-05-03
Abhay Mundepi	2026-05-05
Sanjeev Sangwan	2026-05-06
Yuvraj	2026-05-08
Devansh Thakur	2026-05-10

6 rows in set (0.00 sec)

← || OUTPUT TABLES ||



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-- || Find Doctor-wise Appointments ||

```
SELECT d.name AS doctor_name, COUNT(a.appointment_id) AS total_appointments
FROM Doctors d
LEFT JOIN Appointments a ON d.doctor_id = a.doctor_id
GROUP BY d.name;
```

|| OUTPUT TABLES ||

```
mysql> SELECT d.name AS doctor_name, COUNT(a.appointment_id) AS total_appointments
-> FROM Doctors d
-> LEFT JOIN Appointments a ON d.doctor_id = a.doctor_id
-> GROUP BY d.name;

+-----+-----+
| doctor_name | total_appointments |
+-----+-----+
| Rajveer Singh | 2 |
| Badal Prasad | 2 |
| Shruti Singh | 2 |
| Shiv | 2 |
| Aditya | 2 |
+-----+-----+
5 rows in set (0.00 sec)
```

Conclusion

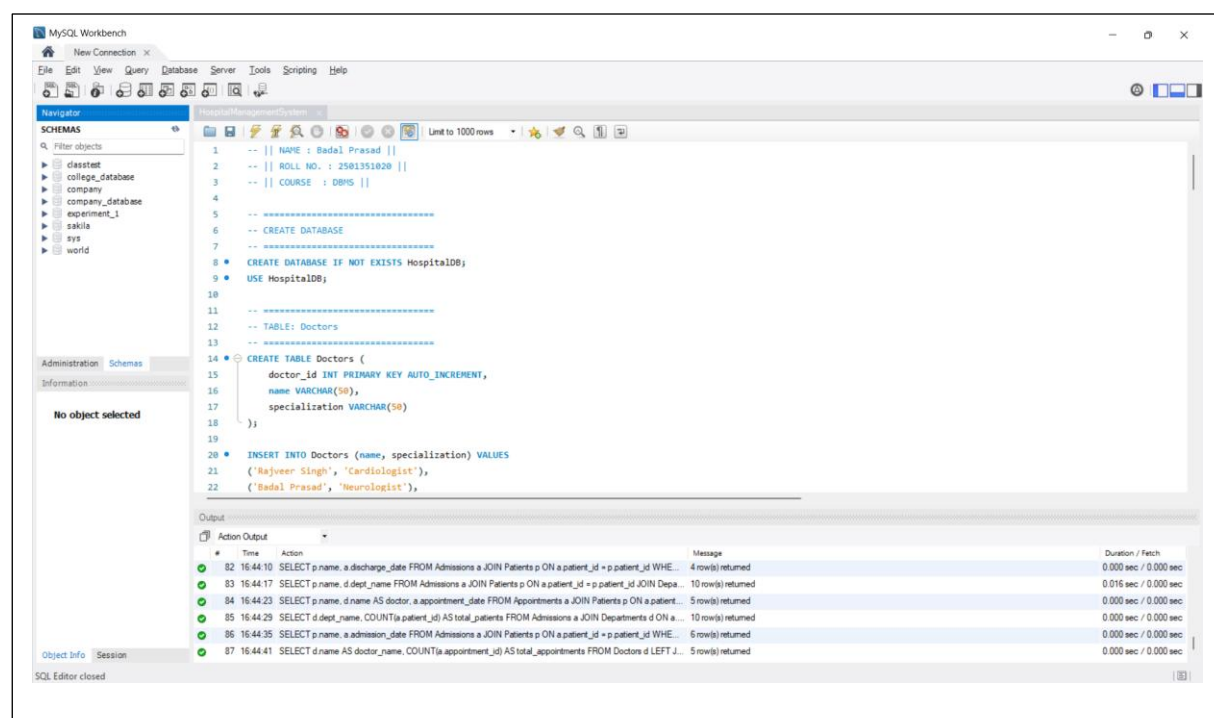
This project successfully demonstrates the implementation of a **Hospital Management System using DBMS concepts.**

Key learnings:

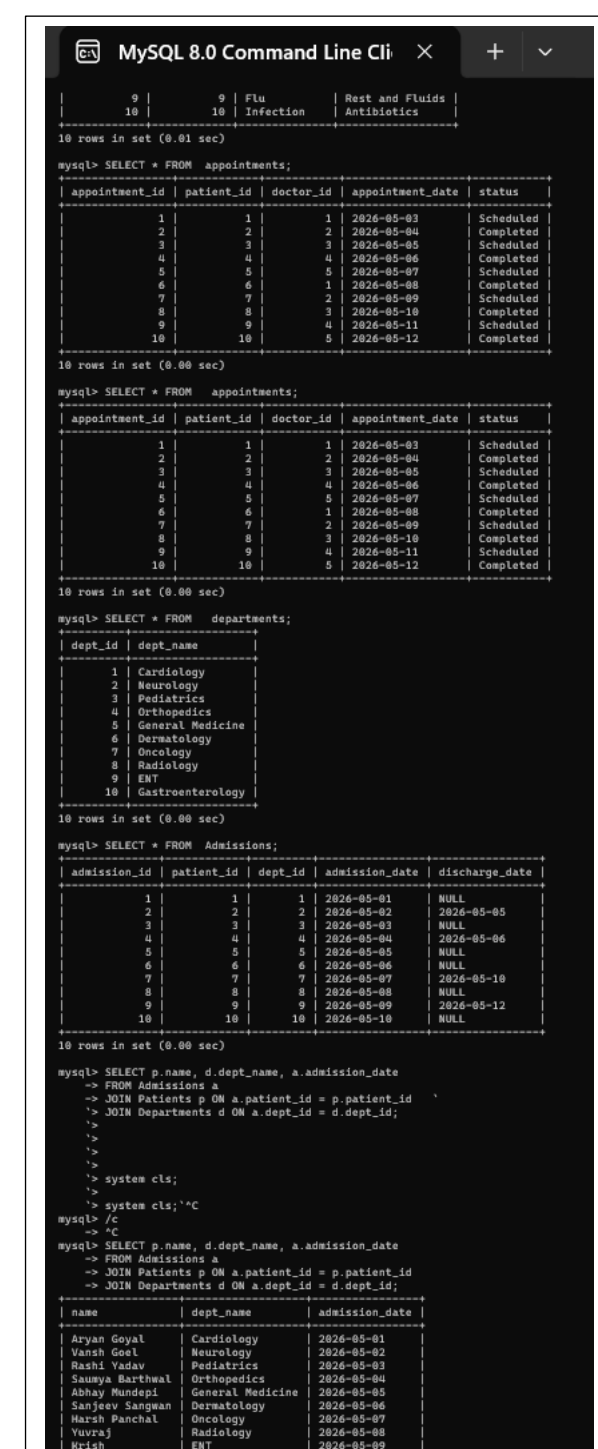
- Designing relational tables
- Using Primary & Foreign Keys
- Writing SQL queries with JOIN, GROUP BY, and WHERE
- Managing real-world hospital data

The system improves **data organization, efficiency, and patient management.**

|| MY SQL ||

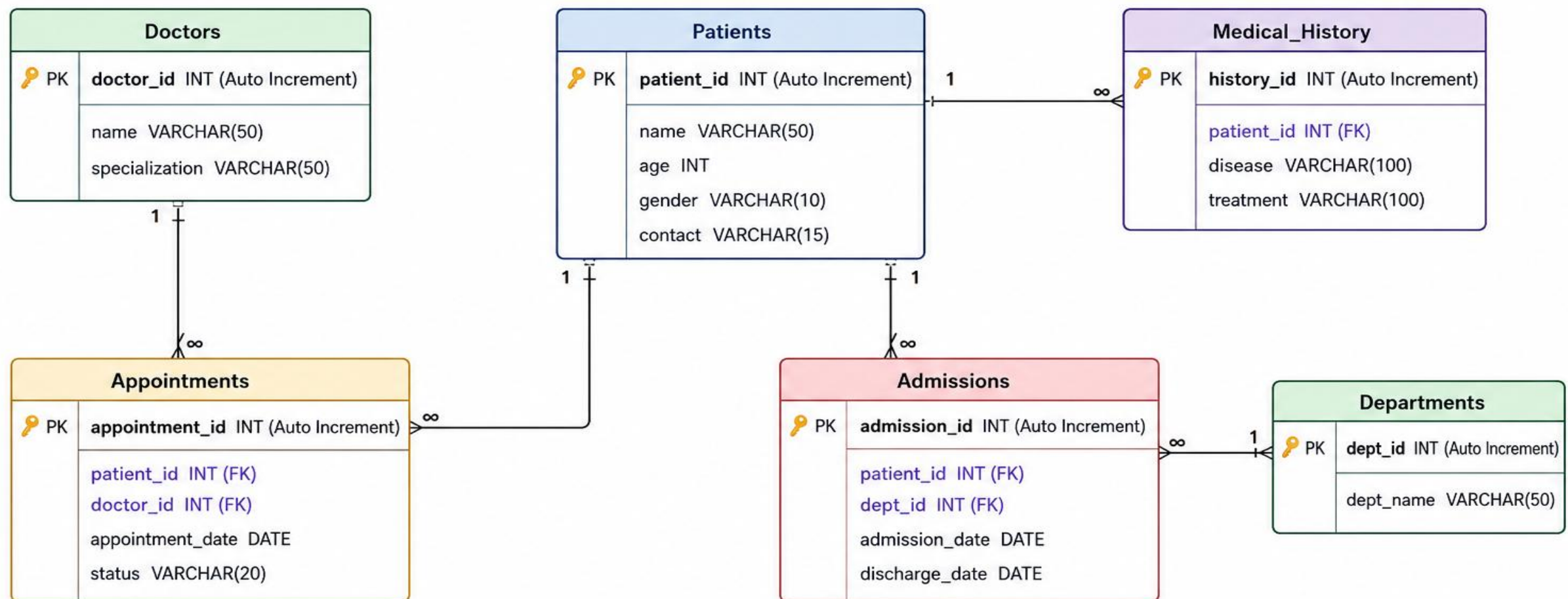


MySQL 8.0 Command Line Client





HOSPITALDB – ER DIAGRAM



LEGEND	
PK	Primary Key
FK	Foreign Key
1	One
∞	Many (Many side)

RELATIONSHIPS SUMMARY	
Patients	1 —< Medical_History
Patients	1 —< Appointments
Doctors	1 —< Appointments
Patients	1 —< Admissions
Departments	1 —< Admissions

DATABASE: HospitalDB	
• 10 Tables	
• Primary Keys (PK) uniquely identify each record.	
• Foreign Keys (FK) enforce relationships between tables.	
• This ER Diagram represents the structure of the Hospital Management System.	





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